

CRISP-DM: Evaluation

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1 Evaluate Results

1.1 Data Mining Results Evaluation

The evaluation starts with a reflection on the results obtained in the previous stage. Each model is evaluated in terms of meeting the data mining success criteria. First, the non-modeling related criteria are evaluated:

- **Feature count:** The project identified 19 developer-level features covering application history, reviews, and temporal patterns, exceeding the required 10. From the business perspective, the data mining stage successfully identified key characteristics of a developer from the market data.
- **Clustering:** K-means clustering ($k = 5$) on the training set achieved a silhouette score of 0.32 (> 0.25). Cluster 0 (Premium Discounters) has a partnership density of 34.9%, well above the 10% threshold. Thus, the clustering criterion is satisfied, and the business received a set of interpretable natural groupings of developers.

Next, the modeling part is assessed:

1. Logistic Regression

Although the logistic regression model nearly exceeds the threshold of separability between top-partnered and non-top-partnered developers (ROC-AUC = 0.803), the model provides a large set of highly interpretable features with reliable probability estimates (well-calibrated, Brier = 0.078, ECE = 0.024). Nevertheless, the final predictions of this model on the test set demonstrate poor precision (40%), which would lead to a potential waste of business resources. Moreover, a large drop in recall on the set of top-partnered emerging developers (from 29% on the test set to 8%) indicates poor transferability to the target audience.

2. Decision Tree

The decision tree achieves good separability (ROC-AUC = 0.822) and provides a transparent set of decision rules. However, the model is poorly calibrated (Brier = 0.161, ECE = 0.254), meaning its probability estimates are not reliable for decision making. While recall on the test set (36%) and on top-partnered emerging developers (33%) are acceptable, precision remains low (39%), which would lead to many false positive recommendations.

3. Random Forest

The base random forest achieves excellent separability (ROC-AUC = 0.882) and good calibration (Brier = 0.070, ECE = 0.068). The model identifies 18 influential features, well above the minimum of three. The model meets the precision target (62%) and recall falls within the desired range (25%). The recall drop on top-partnered emerging developers is 48%, which is below the 50% limit, indicating good transferability.

4. XGBoost

XGBoost performs similarly to random forest in terms of separability (ROC-AUC = 0.881) and calibration (Brier = 0.065, ECE = 0.019). The model identifies 16 important features. Precision (63%) and recall (25%) both meet the criteria. The recall drop on top-partnered emerging developers is 38%, well within the acceptable range.

5. Calibrated Decision Tree

The calibrated decision tree shows improved separability (ROC-AUC = 0.848) and remains interpretable through its tree structure. Calibration significantly enhances the reliability of probability estimates (Brier = 0.071, ECE = 0.010). However, precision remains below the required 60% (47%). Recall on the test set (27%) is adequate, and the recall drop on top-partnered emerging developers is low (21%), demonstrating good transferability. Despite better calibration, the model still does not meet the precision criterion.

6. Calibrated Random Forest

The calibrated random forest maintains high separability (ROC-AUC = 0.882) and achieves excellent calibration (Brier = 0.064, ECE = 0.008). The model still identifies 18 important features. Precision (60%) exactly meets the threshold, and recall (27%) is within the target range. The recall drop on top-partnered emerging developers is 47%, still acceptable. Compared to the base version, precision is slightly lower but still passes the criterion.

1.2 Business Success Criteria Compliance Evaluation

The final evaluation is based on the business success criteria. [Table 1](#) presents the number of emerging non-top developers identified by each model at the probability threshold of 0.7. The business success criterion requires at least five such developers.

Table 1: Emerging non-top developers with predicted probability > 0.7.

Model	Developers found	Meets criterion (≥ 5)
Logistic Regression	26	Yes
Decision Tree	2,238	Yes
Random Forest	48	Yes
XGBoost	0	No
Decision Tree (calibrated)	0	No
Random Forest (calibrated)	3	No

The evaluation below combines the interpretability results from the data mining success criteria with the search performance shown in [Table 1](#) to assess each model’s ability to satisfy the two

business goals: (1) identifying meaningful developer characteristics and (2) producing a reliable scoring system for emerging developers.

1. Logistic Regression

The logistic regression model identifies a large set of interpretable features (17) and meets the search criterion by finding 26 promising emerging developers. However, its poor precision (40%) and substantial recall drop on top-partnered emerging developers (73%) undermine the trustworthiness of its recommendations. While the model fulfills the business requirements in terms of feature discovery and candidate count, the high false positive rate would likely waste business resources in practice.

2. Decision Tree

The base decision tree discovers 11 important features and finds a very large number of candidates (2,238). Nevertheless, the model is poorly calibrated and exhibits low precision (39%), meaning that the vast majority of its recommendations would be false positives. Consequently, despite satisfying the numerical criterion, the model is unsuitable for reliable business use.

3. Random Forest

The base random forest identifies 18 influential features and finds 48 emerging developers with probability above 0.7, well exceeding the required five. The model also demonstrates good precision (62%), acceptable recall (25%), and moderate recall drop on top-partnered emerging developers (48%). Thus, the random forest satisfies both business goals: it provides interpretable characteristics and a practical scoring system.

4. XGBoost

XGBoost identifies 16 important features but fails to find any emerging developer with probability exceeding 0.7. Although the model achieves high precision (63%) and good calibration, its extreme conservatism makes it useless for the business objective of discovering promising young studios. Hence, XGBoost does not meet the business success criteria.

5. Calibrated Decision Tree

Calibration improves the decision tree’s reliability but the model still finds zero candidates above the 0.7 threshold. While it identifies 11 features and shows good transferability, the inability to produce sufficiently confident predictions renders it ineffective for the scoring task.

6. Calibrated Random Forest

The calibrated random forest maintains excellent interpretability (18 features) but finds only three emerging developers above 0.7, two short of the required five. Despite meeting the precision threshold (60%) and calibration targets, the model fails the primary business success criterion.

1.3 Model Recommendation

Based on this evaluation, the **base random forest model** is the only model that simultaneously provides a rich set of interpretable characteristics and identifies a sufficient number of promising emerging developers (≥ 5) with adequate precision. Therefore, this model is recommended for

deployment in the publisher’s developer scouting process. With the reported stages and well documented and versioned code, the next iterations can be automated in the deployment stage as well.

1.4 Understanding of Model Performance

Interpretation

Having established that the uncalibrated Random Forest satisfies both business success criteria, this section reports the specific developer characteristics identified as most influential. [Figure 1](#) displays the mean absolute SHAP values for the Random Forest model, ranking features by their contribution to predictions.

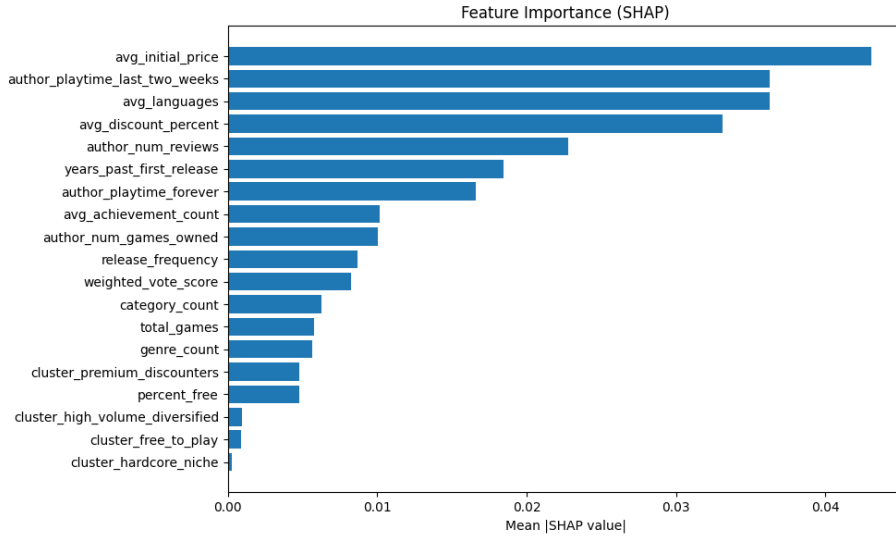


Figure 1: Mean absolute SHAP values for the Random Forest model (test set).

The final set of **top ten predictive features comprises ten developer-level metrics**: average initial price, recent player playtime (minutes in the last two weeks), average supported languages per game, average discount percentage, average number of reviews written per player, developer experience measured as years since first release, total player playtime (minutes), average achievements per game, average number of games owned per player, and game release frequency (games per year). All features are aggregated per developer using either per-game averages or global averages.

Among the cluster indicators, the `cluster_premium_discounters` (cluster 0) feature exhibits the highest importance, while the other cluster labels appear at the bottom of the ranking. This indicates that belonging to the premium discounter segment is a strong positive signal for top-partnered status, whereas membership in other clusters (hardcore niche, high-volume diversified, free-to-play) contributes little additional information beyond the continuous features.

These findings confirm that the publisher can focus on a manageable set of observable developer attributes: notably pricing strategy, recent player engagement, localization effort, and portfolio di-

versity. The prominence of the premium discounter cluster further suggests that developers offering moderately priced games with frequent discounts are particularly attractive to top publishers.

Emergent Developers Ranking

To understand why the Random Forest model assigns high probabilities to certain emerging developers, SHAP waterfall analysis was performed on the top 10 ranked candidates (predicted probability > 0.7). Figure 2 indicates the results for the selected candidates.



Figure 2: SHAP Waterfall Plots

The analysis reveals two distinct archetypes of promising developers, each characterized by a different set of positive drivers.

1. Archetype I: Premium Engagement Studios

Developers 1, 2, 3, 8, and 10 exhibit a profile dominated by `avg_initial_price`, `author_playtime_last_two_weeks`, and `avg_languages`. For these studios, the model's confidence stems from premium pricing, strong recent player engagement, and broad language support. Examples include Developer 1, where these three features contribute +0.193, +0.145, and +0.096

respectively, while `avg_discount_percent` plays a negligible or slightly negative role. This archetype corresponds to high-quality, internationally oriented developers that attract active players.

2. *Archetype II: Discount-Driven Premium Discounters*

Developers 4, 5, 6, 7, and 9 share a different pattern: the largest positive contributions come from `avg_discount_percent` and the `cluster_premium_discounters` indicator. For instance, Developer 4 receives +0.305 from discount percentage and +0.118 from belonging to the premium discounter cluster. In this group, `avg_initial_price` also contributes positively but to a lesser extent (e.g., +0.051 for Developer 4). These developers are characterised by frequent discounts and the specific segment identified earlier as the most partnership-dense cluster.

3. *Consistent Negative Drivers*

Across almost all top-ranked developers, `years_past_first_release` exerts a small negative influence (typically -0.01 to -0.03). This reflects the business expectation that younger studios (with fewer years since first release) are the target of the scouting system. Occasional negative contributions from `author_num_reviews` or `author_num_games_owned` (e.g., Developer 4, -0.027) indicate that extremely low reviewer activity can slightly lower confidence.

The waterfall analysis confirms that the Random Forest model relies on economically interpretable features and identifies two viable pathways to high predicted probability: (i) premium pricing with high engagement and language coverage, and (ii) aggressive discounting combined with membership in the premium discounter cluster. Both archetypes satisfy the business requirement of providing actionable, transparent characteristics for scouting promising emerging developers.

2 Review Process

The data mining process was reviewed for correctness, data leakage, and adherence to assumptions. All splits (mature/emerging, train/test) were properly stratified, and clustering was fitted only on the training set. The analysis explicitly acknowledges that correlations do not imply causality, and the business constraints (top 100 publishers by volume, emerging definition of three years) were respected.

Nevertheless, several problems emerged during the project and were addressed correctly. First, the dataset contained noise in the form of missing price information: many free games had missing values in price columns instead of being explicitly set to zero, requiring targeted imputation. Second, defining the train/test split proved challenging: the initial plan used a simple stratified split, but after recognizing that emerging developers (first release ≤ 3 years) have unreliable negative labels, the split had to be redesigned to remove all emerging developers from the training set and reserve them solely for final business evaluation. Third, the importance of model calibration became apparent only after examining the probability distributions of uncalibrated models; calibration was therefore added as a primary data mining goal to ensure that predicted probabilities could be meaningfully interpreted with an absolute threshold (e.g., 0.7) for business decision making.

No critical steps were omitted; however, the recall drop on top-partnered emerging developers (48% for the chosen model) suggests that transferability could be improved by collecting more positive examples of young successful developers. The precision threshold (60%) was met, but further

calibration might slightly increase false positives. Moreover, since only one out of three promising models satisfied the final business success criteria, the evaluation strategy can be enhanced to detect poor performance in terms of identifying emerging developers on the early stages. Overall, the process is reproducible and the results are reliable for the stated business objectives.

3 Determine Next Steps

Based on the current results, the project meets the business success criteria and can be deployed as is. However, several improvements are possible: collecting more positive examples of young successful developers would strengthen the training set, modeling can be enriched with additional models (SVM, neural networks), and the evaluation strategy could be enhanced to detect poor performance in identifying promising emerging developers at an early stage. No additional CRISP-DM iteration is strictly required because the business criteria are already satisfied, but a follow-up iteration could incorporate the suggested improvements. More data is not necessary for deployment, but future iterations should attempt to obtain additional observations of young top-partnered developers. The current model and ranking system are ready for deployment. The main recommendation is to start contacting the found emerging developers to initiate negotiation processes, using the model's interpretation reports (e.g., SHAP waterfalls) to discuss risks and opportunities with domain experts.